

Evaluation Scenarios and Data Collection Strategy

Cross-Chain Content Rights Management Service

This document delineates the evaluation scenarios and data collection strategy for the CCRMS framework, aligning with the measurement-focused research objectives (RO1–RO6) and the Key Performance Indicators (KPIs) established in the Methodology chapter. The evaluation is structured to document attainable outcomes across technical, economic, and user-centric dimensions rather than to assess against predetermined benchmarks.

1 Evaluation Framework Overview

1.1 Evaluation Philosophy

The evaluation adopts an **exploratory, measurement-focused approach** wherein the primary objective is to:

1. **Document** what levels of performance, efficiency, and usability the CCRMS framework achieves
2. **Compare** measured results against established baselines (Ethereum L2s, centralised DRM, bridge-based solutions)
3. **Analyse** trade-offs between competing objectives (e.g., decentralisation vs. latency)
4. **Identify** limitations, bottlenecks, and areas for future improvement

This method ensures a valid academic contribution, irrespective of particular results achieved.

1.2 Mapping to Research Objectives

Table 1: Evaluation Scenarios Mapped to Research Objectives

Research Objective	Evaluation Focus	Primary Scenarios
RO1: XCM extensions for recurring transfers	Atomic success rates, cross-chain finality times	ES-01, ES-02, ES-03
RO2: Unified ink! rights token	Access verification latency	ES-04, ES-05, ES-06
RO3: Composable rights pallet	Feasibility and trade-offs of unification	ES-07, ES-08
RO4: Benchmark against alternatives	TPS, costs, revenue retention comparison	ES-09, ES-10, ES-11
RO5: Zero-knowledge proof hooks	Implementation feasibility	ES-12
RO6: Creator savings quantification	Monte-Carlo simulation results	ES-13

2 Evaluation Scenarios

2.1 Category A: Cross-Chain Functionality (RO1)

2.1.1 ES-01: Basic XCM Transfer Success Rate

Objective: Assess the success rate and finality duration of XCM messages that transmit rights metadata between parachains.

Scenario Description:

- Initiate 100 XCM transfers from CCRMS parachain to Asset Hub
- Each transfer carries a simulated rights token with metadata (expiration, royalty config)
- Vary payload sizes: minimal (256 bytes), standard (1 KB), maximum (4 KB)

Table 2: ES-01 Test Parameters

Parameter	Values
Number of transfers	100 per payload size (300 total)
Payload sizes	256 B, 1 KB, 4 KB
Network conditions	Normal, congested (simulated)
Repetitions	3 runs per configuration

Data to Collect: Success/failure status for each transfer; time from send to confirmation (finality time); error types for failed transfers; gas/weight consumed per transfer.

Expected Outputs: Success rate percentage per configuration; mean, median, and standard deviation of finality times; failure mode classification.

2.1.2 ES-02: Scheduled XCM Renewal Accuracy

Objective: Evaluate the reliability and timing accuracy of scheduled XCM messages for subscription auto-renewal.

Scenario Description:

- Schedule 50 subscription renewals at specific future block heights
- Block height intervals: 10 blocks, 100 blocks, 1000 blocks ahead
- Measure actual execution time vs. scheduled time

Table 3: ES-02 Test Parameters

Parameter	Values
Number of scheduled renewals	50 per interval (150 total)
Scheduling intervals	10, 100, 1000 blocks
Pre-authorised balance states	Sufficient, insufficient, exact

Data to Collect: Scheduled block height vs. actual execution block; timing deviation (blocks early/late); success rate based on balance availability; behaviour during chain congestion.

Expected Outputs: Scheduling accuracy metrics (deviation distribution); renewal success rate by balance state; identified edge cases and failure modes.

2.1.3 ES-03: Cross-Chain Royalty Distribution

Objective: Measure the reliability and latency of multi-recipient royalty distributions across chains.

Scenario Description:

- Simulate secondary sale triggering royalty distribution
- Recipients distributed across: CCRMS parachain (2), Asset Hub (2), external chain simulation (1)
- Vary royalty split complexity: 2, 5, 10 recipients

Table 4: ES-03 Test Parameters

Parameter	Values
Number of test sales	30 per split configuration (90 total)
Recipient counts	2, 5, 10
Recipient locations	Mixed (local + cross-chain)

Data to Collect: Time from sale to all royalties received; per-recipient delivery success/failure; total gas cost for distribution; partial failure handling behaviour.

Expected Outputs: End-to-end royalty distribution time by complexity; success rate per recipient location type; cost scaling analysis.

2.2 Category B: Access Verification Performance (RO2)

2.2.1 ES-04: Subscription Access Verification Latency

Objective: Measure end-to-end latency for subscription-based access verification under varying conditions.

Table 5: ES-04 Test Parameters

Parameter	Values
Number of requests	500 per state (1500 total)
Subscription states	Active (>7 days), Near-expiry (<24 hours), Expired
Concurrent requests	1, 10, 50, 100 simultaneous

Data to Collect: Request-to-response latency (milliseconds); breakdown: network, on-chain query, response formatting; throughput under concurrent load; error rates under load.

Expected Outputs: Latency distribution by subscription state; latency vs. concurrency curves; maximum sustainable concurrent requests.

2.2.2 ES-05: PPV Counter Verification and Decrement

Objective: Measure latency and accuracy of PPV counter operations (check + decrement).

Table 6: ES-05 Test Parameters

Parameter	Values
Number of view requests	200 per counter state (600 total)
Counter states	Full (10), Partial (5), Final (1), Exhausted (0)
Request patterns	Sequential, burst (10 rapid requests)

Data to Collect: Verification + decrement latency; counter accuracy (no double-spend, no over-decrement); behaviour at boundary (counter = 1 $\rightarrow$ 0); race condition handling under burst.

Expected Outputs: Latency distribution for PPV operations; counter integrity verification results; concurrent access handling analysis.

2.2.3 ES-06: Ownership Verification with Perpetual Subscription

Objective: Verify that ownership check correctly returns perpetual subscription status without separate queries.

Table 7: ES-06 Test Parameters

Parameter	Values
Number of verifications	300
Ownership states	Owner, non-owner, bridged owner
Query methods	Unified query, separate queries (for comparison)

Expected Outputs: Latency comparison: unified vs. separate approach; storage efficiency metrics; validation of unified rights token concept.

2.3 Category C: Monetisation Model Integration (RO3)

2.3.1 ES-07: Model Transition Scenarios

Objective: Evaluate the feasibility and user experience of transitioning between monetisation models.

Table 8: ES-07 Expected Transition Behaviours

Transition	Expected Behaviour
PPV $\rightarrow$ Subscription	Retain remaining views + add subscription
Subscription $\rightarrow$ Ownership	Convert to perpetual, cancel scheduled renewals
Expired Subscription $\rightarrow$ PPV	Issue View Pack, retain expired sub record
PPV $\rightarrow$ Ownership	Retain remaining views + add perpetual access

Data to Collect: Transaction success/failure for each transition; state consistency before/after transition; user-facing latency for upgrade operations; edge case handling (e.g., mid-renewal upgrade).

2.3.2 ES-08: Bundled Access Verification

Objective: Test scenarios where users hold multiple access types simultaneously.

Table 9: ES-08 Access Resolution Test Cases

Content Requested	User Rights	Expected Resolution
Content A	Subscription (active)	Grant via subscription
Content B	PPV (5 views)	Grant via PPV, decrement
Content C	Ownership	Grant via perpetual
Content D	None	Deny access
Content A	Subscription (expired) + PPV available	Prompt renewal or use PPV

Expected Outputs: Access resolution accuracy rate; priority handling documentation; complex state management analysis.

2.4 Category D: Scalability and Performance (RO4)

2.4.1 ES-09: Transaction Throughput Under Load

Objective: Measure maximum sustainable TPS for core operations on the CCRMS parachain.

Table 10: ES-09 Test Parameters

Parameter	Values
Load levels (TPS target)	10, 50, 100, 200, 500, 1000
Test duration per level	5 minutes sustained
Operation mix	60% access verifications, 25% PPV purchases, 10% subscriptions, 5% ownership transfers
Node configurations	5, 10, 20, 50 validator nodes

Expected Outputs: TPS capacity curve by node configuration; latency degradation analysis; bottleneck identification; optimal operating parameters.

2.4.2 ES-10: Comparative Benchmark: Ethereum L2

Objective: Compare CCRMS performance against equivalent operations on Ethereum L2 (e.g., Arbitrum, Optimism).

Table 11: ES-10 Comparison Metrics

Metric	CCRMS Measurement	Ethereum L2 Measurement
TPS (PPV operations)	From ES-09	Equivalent test
Transaction cost	DOT/KSM equivalent	ETH gas cost
Finality time	Block confirmation	L2 + L1 finality
Cross-chain transfer time	XCM to Asset Hub	Bridge to L1/other L2

Expected Outputs: Comparative performance table; cost efficiency analysis; trade-off discussion for thesis.

2.4.3 ES-11: Comparative Benchmark: Centralised DRM

Objective: Compare CCRMS latency and costs against centralised DRM baseline.

Table 12: ES-11 Comparison Metrics

Metric	CCRMS	Centralised Baseline
Access verification latency	Measured	Lambda cold/warm start
Throughput capacity	Measured TPS	Lambda concurrent executions
Cost per 1000 operations	DOT/KSM fees	AWS pricing
Decentralisation (HHI)	Calculated	N/A (centralised)

Expected Outputs: Latency comparison analysis; total cost of ownership comparison; trade-off matrix (decentralisation vs. performance).

2.5 Category E: Security and Reliability (RO4 continued)

2.5.1 ES-12: Zero-Knowledge Proof Feasibility (RO5)

Objective: Assess the feasibility of integrating ZK proof hooks for selective regulatory disclosure.

Table 13: ES-12 Test Parameters

Parameter	Values
ZK framework	RISC Zero or zkMega (based on availability)
Proof types	Subscription validity, purchase ownership, revenue threshold
Hardware	Standard development machine (baseline)

Data to Collect: Proof generation time; proof verification time (on-chain); proof size (bytes); integration effort estimate (developer hours).

Expected Outputs: Feasibility assessment (viable/not viable/conditionally viable); performance characteristics documentation; recommendations for future implementation.

2.6 Category F: Economic Analysis (RO6)

2.6.1 ES-13: Monte-Carlo Creator Revenue Simulation

Objective: Quantify potential creator savings using the CCRMS framework compared to alternatives.

Table 14: ES-13 Simulation Parameters

Parameter	Distribution
Audience size	Log-normal ($\mu=1000$, $\sigma=2$)
Content price	Uniform (\$1–\$50)
Subscription rate	Beta ($\alpha=2$, $\beta=5$)
PPV conversion	Beta ($\alpha=1$, $\beta=10$)
Secondary sale probability	Exponential ($\lambda=0.1$)

Table 15: Platform Fee Structures

Platform Type	Creator Retention
Centralised (YouTube, Spotify)	55–70%
Existing Web3 (OpenSea, etc.)	85–92.5%
Bridge-based solutions	85–92%
CCRMS (estimated)	95–99% (to be measured)

Expected Outputs: Expected creator savings distribution; confidence intervals for savings estimates; scenario analysis (best/worst/typical cases).

3 Data Collection Strategy

3.1 Quantitative Data Collection

3.1.1 On-Chain Metrics

Collection Method: Substrate telemetry + custom event logging

Tools:

- Prometheus for metrics aggregation
- Grafana for visualisation and alerting
- SubQuery for historical event indexing
- Custom Rust scripts using subxt for transaction submission

Table 16: On-Chain Metrics Collection

Metric Category	Specific Metrics	Collection Frequency
Transaction Performance	TPS, latency, gas used	Per-block
XCM Operations	Success rate, finality time, payload size	Per-message
State Changes	NFT mints, transfers, burns, counter updates	Per-event
Resource Utilisation	Block weight, storage growth	Per-block

Data Schema:

```
{
  "timestamp": "ISO-8601",
  "block_number": integer,
  "event_type": string,
  "operation": string,
  "latency_ms": float,
  "gas_used": integer,
  "success": boolean,
  "error_code": string | null,
  "metadata": object
}
```

3.1.2 Off-Chain Metrics**Collection Method:** Application-level instrumentation

Table 17: Off-Chain Metrics Collection

Metric Category	Specific Metrics	Collection Frequency
API Performance	Request latency, throughput	Per-request
User Sessions	Session duration, actions per session	Per-session
Error Rates	Error types, frequency, recovery time	Per-error

3.1.3 Comparative Baseline Data**Ethereum L2 Data:**

- Deploy equivalent contracts to Arbitrum Sepolia / Optimism Goerli
- Use Hardhat for deployment and testing
- Etherscan API for transaction data

Centralised Baseline Data:

- AWS CloudWatch metrics

- Lambda execution logs
- DynamoDB consumed capacity units

3.2 Qualitative Data Collection

3.2.1 Expert Interviews

Participants: 5–10 blockchain developers/researchers with Polkadot/Kusama experience

Recruitment: Polkadot forums, academic networks, Web3 developer communities

Interview Protocol:

- Semi-structured format (45–60 minutes)
- Topics: Architecture review, feasibility assessment, improvement suggestions
- Recording: Audio with consent, transcribed for analysis

Sample Questions:

1. How does the RMRK 2.0 + scheduled XCM approach compare to alternatives you've seen?
2. What potential issues do you foresee with the unified rights token model?
3. How would you assess the production-readiness of this architecture?

3.2.2 User Testing Sessions

Participants: 20–30 users (mix of content creators and consumers)

Session Structure:

1. Pre-test questionnaire (demographics, Web3 experience)
2. Guided task completion (subscribe, PPV purchase, ownership transfer)
3. Think-aloud protocol during tasks
4. Post-test questionnaire (usability, satisfaction)
5. Semi-structured debrief interview

Table 18: User Testing Tasks

Task ID	Description	Monetisation Model
UT-01	Subscribe to content for 30 days	Subscription
UT-02	Purchase 5-view pack and consume 2 views	PPV
UT-03	Buy content permanently	Purchase
UT-04	Transfer owned content to different wallet	Ownership transfer
UT-05	Upgrade from subscription to ownership	Model transition

3.2.3 Usability Questionnaires

Instrument: System Usability Scale (SUS) + custom questions

SUS Questions: Standard 10-item scale, scored 0–100, industry benchmark ~68.

Custom Questions (5-point Likert scale):

1. The process of subscribing to content was straightforward.
2. I understood how my PPV views were being tracked.
3. I felt confident that my ownership rights were secure.
4. The cross-chain transfer process was clear.
5. I would use this system for real content purchases.

3.3 Data Collection Timeline

Table 19: Data Collection Timeline

Week	Activities	Data Types
17	Testnet setup, baseline configuration, ES-01 to ES-03; centralised baseline deployment	On-chain XCM metrics; comparative baseline data
18	ES-04 to ES-08, user recruitment; user testing sessions (n=10)	Access verification metrics; qualitative usability data
19	ES-09 to ES-11, expert interviews; user testing sessions (n=10–20)	Scalability metrics; qualitative expert feedback; additional usability data
20	ES-12, ES-13, data consolidation; data analysis and preliminary findings	ZK feasibility; simulation results; all data types

3.4 Data Storage and Management

Storage Infrastructure:

- Raw data: Encrypted cloud storage (institutional account)
- Processed data: Local analysis environment
- Code and scripts: GitHub repository (public, MIT license)

Data Retention:

- Quantitative data: Retained indefinitely for reproducibility
- Qualitative data: Anonymised transcripts retained; audio deleted after transcription
- User data: Anonymised within 30 days of collection

4 Analysis Approach

4.1 Quantitative Analysis

Descriptive Statistics: Central tendency (mean, median, mode) for all metrics; dispersion (standard deviation, IQR, range); distribution visualisation (histograms, box plots).

Comparative Analysis: Paired comparisons (CCRMS vs. Ethereum L2, CCRMS vs. centralised); statistical significance testing where appropriate (t-tests, Mann-Whitney U); effect size calculation (Cohen's d).

Performance Modelling: Regression analysis for latency vs. load relationships; capacity modelling using queuing theory; cost projection models.

4.2 Qualitative Analysis

Thematic Analysis: Open coding of interview transcripts; axial coding to identify relationships; selective coding for core themes.

Usability Analysis: SUS score calculation and benchmarking; task completion rate analysis; error classification and frequency.

4.3 Mixed-Methods Integration

Table 20: Mixed-Methods Integration Points

Quantitative Finding	Qualitative Validation
Access latency measurements	User perception of responsiveness
XCM success rates	Expert assessment of reliability
Cost metrics	Creator willingness to adopt

5 Ethical Considerations

5.1 Participant Protection

- Informed consent obtained from all interview and testing participants
- Right to withdraw at any time without consequence
- No collection of personally identifiable information beyond demographics
- Data anonymisation before analysis

5.2 Research Integrity

- All test scenarios documented and reproducible
- Raw data preserved for verification
- Negative or unexpected results reported transparently

- Limitations explicitly acknowledged

5.3 Institutional Compliance

- Ethics approval obtained from University of Malta (if required)
 - GDPR compliance for any EU participant data
 - Open-source code release for community verification
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6 Summary

This evaluation strategy provides a comprehensive framework for measuring the CCRMS framework’s performance, usability, and economic viability. The 13 evaluation scenarios cover all six research objectives, while the mixed-methods data collection strategy ensures both rigorous quantitative measurement and rich qualitative insight.

Key Deliverables from Evaluation:

1. Performance metrics across all KPI categories
2. Comparative analysis against Ethereum L2 and centralised alternatives
3. Feasibility assessment for ZK proof integration
4. Monte-Carlo simulation results for creator savings
5. Usability scores and user feedback synthesis
6. Expert validation of architectural decisions

The measurement-focused approach ensures that regardless of specific outcomes, the evaluation produces meaningful academic contribution through documented findings, comparative analysis, and actionable recommendations.

This document satisfies the Phase 2 Week 9–10 deliverable: “Design evaluation scenarios and data collection strategy.” It is prepared for integration into the thesis Methodology and Evaluation chapters.

Phase 5 Evaluation Mapping

The following table maps the planned evaluation scenarios to the actual testing performed in Phase 5 (Weeks 17–20).

Table 21: Planned vs Actual Evaluation Coverage

Scenario	Planned	Actual Test	Coverage
ES-01: XCM Transfer Success Rate	100 transfers, 3 payload sizes	xcm-latency, 9 cross-chain operations, 100% success	Partially covered — fewer iterations but all succeeded
ES-02: Scheduled XCM Renewal	50 scheduled renewals	Unit tests: auto_renew_policy_expired_sub validates on-chain scheduler. XCM scheduled renewal not tested (replaced by on_initialize pattern)	Covered differently
ES-03: Bridge Latency	Snowbridge round-trip	Snowbridge E2E: 2 ETH bridged Ethereum → AssetHub via full relay pipeline	Covered — exceeded plan with full v1 bridge
ES-04: Access Verification Latency	<1 second target	local-throughput measured at ~6,000ms (1 block). Documented as performance trade-off	Covered: target not met, documented in thesis
ES-05–06: Rights Token Operations	RMRK-based operations	local-throughput + stress-test.mjs: all 6 extrinsic types benchmarked at batch 1–1000	Fully covered
ES-07–08: Composable Rights	Feasibility	56 unit tests + XCM simulator + Zombienet E2E validate unified rights model	Fully covered
ES-09–11: Benchmark vs Alternatives	¹³ Ethereum L2, centralized DRM	centralized-Ex-press+SQLite	Partially covered — benchmark/: centralized

Actual Deliverables Produced (vs Planned)

Table 22: Planned vs Actual Deliverables

Planned	Actual
Performance metrics across all KPIs	11 KPIs measured, 10/11 targets met (Chapter 6 Table 6.2)
Comparative analysis vs alternatives	Centralized benchmark: 270× TPS, 67,000× latency ratios
ZK proof feasibility	Deferred to future work
Monte-Carlo simulation	Deferred to future work
Usability scores	Not in scope (no user study conducted)
Expert validation	Not Complete